Robustness and Generalization Limits of Deep Ensembles across Multi-Center Clinical Imaging

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Abstract
While recent machine learning literature reports stellar classification accuracy on curated research datasets, cross-center clinical transportability remains an open problem. We conduct a rigorous multi-institution study evaluating whether deep architectures and ensemble methods consistently maintain classification accuracy across diverse hospital acquisition pipelines without site-specific recalibration.

1. Introduction
The translation of medical imaging algorithms from bench to bedside is frequently stymied by out-of-distribution performance drops. Variations in scanner hardware manufacturers, radiologic technician positioning, contrast agent timing, and patient demographics introduce systemic domain shifts. A technique that demonstrates state-of-the-art accuracy in an originating academic medical center may degrade substantially when deployed in a community hospital.

2. Multi-Center Study Protocol
We compiled 14,800 medical imaging studies across four geographically separate hospital networks:
- Hospital A (Originating center): Academic hospital utilizing GE Healthcare 64-slice CT and digital radiography.
- Hospital B: Regional medical center using Siemens Somatom scanners.
- Hospital C: Urban trauma center with Philips Brilliance systems and high trauma prevalence.
- Hospital D: Rural ambulatory clinic with portable digital X-ray equipment.

Evaluation Metrics:
Performance was measured using Area Under the ROC Curve (AUC), Expected Calibration Error (ECE), and worst-case subgroup sensitivity across diagnostic indications.

3. Cross-Site Performance Disagreements
Our empirical findings reveal significant domain sensitivity:
While the proposed ensemble architecture achieved an in-distribution AUC of 0.931 on Hospital A data, performance dropped to 0.824 on Hospital C and 0.795 on Hospital D.

The technique does not consistently improve classification accuracy across heterogeneous clinical centers unless explicit domain adaptation or scanner normalization is applied.

Contrary to claims that novel transformer architectures intrinsically resolve domain shifts, our evidence demonstrates that deep model predictions remain vulnerable to scanner manufacturer variations, slice thickness discrepancies, and institutional patient mix differences.

4. Methodological Conclusions and Recommendations
Our analysis underscores the following boundaries:

1. Calibration Drift: Even when discrimination remained acceptable, model confidence scores drifted significantly; Expected Calibration Error increased from 0.042 (Hospital A) to 0.168 (Hospital C), producing overconfident incorrect classifications.

2. Necessity of Standardized Acquisition: Algorithmic improvements alone cannot overcome raw acquisition discrepancies; institutional DICOM standardization protocols are essential prerequisites for reliable clinical AI.

3. Verification Requirement: Independent multi-center validation must be required by regulatory agencies prior to clinical clearance.